

MH1101 Tutorial 2 (Week 3)

Solution

Reference: Sections 1.3 1.4, 1.5 (Lecture Notes)

1. Use the Fundamental Theorem of Calculus Part 1 to find the derivative of the following functions:

$$(i) \quad g(x) = \int_1^x \cos(t^2) dt$$

$$(ii) \quad F(s) = \int_s^0 \sqrt{1 + \sec t} dt$$

$$(iii) \quad R(y) = \int_{2y}^{3y} \frac{u^2 - 1}{u^2 + 1} du$$

Solution

(i) By FTC Part 1,

$$\begin{aligned} g'(x) &= \frac{d}{dx} \int_1^x \cos(t^2) dt \\ &= \cos(x^2). \end{aligned}$$

(ii) By FTC Part 1,

$$\begin{aligned} F'(s) &= \frac{d}{ds} \int_s^0 \sqrt{1 + \sec t} dt \\ &= \frac{d}{ds} \left(- \int_0^s \sqrt{1 + \sec t} dt \right) \\ &= - \frac{d}{ds} \int_0^s \sqrt{1 + \sec t} dt \\ &= -\sqrt{1 + \sec s}. \end{aligned}$$

(iii) Rewrite $R(y)$ as follows:

$$\begin{aligned} R(y) &= \int_{2y}^{3y} \frac{u^2 - 1}{u^2 + 1} du \\ &= \int_{2y}^0 \frac{u^2 - 1}{u^2 + 1} du + \int_0^{3y} \frac{u^2 - 1}{u^2 + 1} du \\ &= - \int_0^{2y} \frac{u^2 - 1}{u^2 + 1} du + \int_0^{3y} \frac{u^2 - 1}{u^2 + 1} du \end{aligned}$$

Note: Instead of 0, we can use any fixed real number a so that

$$R(y) = \int_{2y}^a \frac{u^2 - 1}{u^2 + 1} du + \int_a^{3y} \frac{u^2 - 1}{u^2 + 1} du$$

since $R(y)$ is defined (and is continuous) for all y .

Next, we cannot apply FTC Part 1 directly. We first need to make the following substitutions for the two integrals on the right:

$$s = 2y, \quad t = 3y.$$

By FTC Part 1 and Chain Rule,

$$\begin{aligned} R'(y) &= \left(-\frac{d}{ds} \int_0^s \frac{u^2 - 1}{u^2 + 1} du \right) \cdot \frac{ds}{dy} + \left(\frac{d}{dt} \int_0^t \frac{u^2 - 1}{u^2 + 1} du \right) \cdot \frac{dt}{dy} \\ &= \left(-\frac{s^2 - 1}{s^2 + 1} \right) (2) + \left(\frac{t^2 - 1}{t^2 + 1} \right) (3) \\ &= 2 \left(-\frac{4y^2 - 1}{4y^2 + 1} \right) + 3 \left(\frac{9y^2 - 1}{9y^2 + 1} \right). \end{aligned}$$

□

2. Evaluate the following definite integral:

$$(i) \int_{-1}^2 (x - 2|x|) dx$$

(ii) $\int_{-2}^2 f(x) dx$, where

$$f(x) = \begin{cases} x^3 & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Solution

(i) Since the expression $|x|$ appears in the function, we split the integral based on the parity of x : if $x \geq 0$, then $|x| = x$; if $x < 0$, then $|x| = -x$. So

$$\begin{aligned} \int_{-1}^2 (x - 2|x|) dx &= \int_{-1}^0 (x - 2|x|) dx + \int_0^2 (x - 2|x|) dx \\ &= \int_{-1}^0 (x - 2(-x)) dx + \int_0^2 (x - 2x) dx \\ &= 3 \int_{-1}^0 x dx - \int_0^2 x dx \\ &= 3 \left[\frac{x^2}{2} \right]_{-1}^0 - \left[\frac{x^2}{2} \right]_0^2 \\ &= -\frac{3}{2} - \frac{4}{2} = -\frac{7}{2}. \end{aligned}$$

(ii) Since $f(x)$ is defined by two branches, we have to split the integral based on these branches:

$$\begin{aligned} \int_{-2}^2 f(x) dx &= \int_{-2}^0 -x dx + \int_0^2 x^3 dx \\ &= \left[-\frac{x^2}{2} \right]_{-2}^0 + \left[\frac{x^4}{4} \right]_0^2 \\ &= 2 + 4 = 6. \end{aligned}$$

□

3. Express the given indefinite integrals in terms of the function f :

$$(i) \int f'(5x) dx$$

$$(ii) \int x f'(3x^2) dx$$

Solution Throughout, C denotes a constant.

(i) Let $u = 5x$. Then $du = 5 dx \implies dx = \frac{1}{5} du$. An antiderivative of $f'(u)$ is just $f(u)$. So

$$\int f'(5x) dx = \int f'(u) \frac{1}{5} du = \frac{1}{5} \int f'(u) du = \frac{1}{5} f(u) + C = \frac{1}{5} f(5x) + C.$$

(ii) Let $u = 3x^2$. Then $du = 6x dx \implies x dx = \frac{1}{6} du$. An antiderivative of $f'(u)$ is just $f(u)$. So

$$\int x f'(3x^2) dx = \int f'(u) \frac{1}{6} du = \frac{1}{6} \int f'(u) du = \frac{1}{6} f(u) + C = \frac{1}{6} f(3x^2) + C.$$

□

4. Evaluate the integral by making a suitable substitution.

$$(i) \int x \sqrt{1-x^2} dx$$

$$(iv) \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

$$(ii) \int x^2 \sin(x^3) dx$$

$$(v) \int \frac{1}{\cos^2 t \sqrt{1 + \tan t}} dt$$

$$(iii) \int_{-5/2}^{-2} x(2x+5)^8 dx$$

$$(vi) \int_{\pi/3}^{2\pi/3} \csc^2 \left(\frac{t}{2} \right) dt$$

Solution Throughout, C denotes a constant. For antiderivatives of some functions, please refer to the table in Section 1.1. of Chapter 1.

(i) Let $u = 1 - x^2$. Then $du = -2x dx \implies x dx = -\frac{1}{2} du$. So

$$\int x \sqrt{1-x^2} dx = -\frac{1}{2} \int u^{1/2} du = -\frac{1}{2} \frac{u^{3/2}}{3/2} + C = -\frac{1}{3} (1-x^2)^{3/2} + C.$$

(ii) Let $u = x^3$. Then $du = 3x^2 dx \implies x^2 dx = \frac{1}{3} du$. So

$$\int x^2 \sin(x^3) dx = \frac{1}{3} \int \sin u du = -\frac{1}{3} \cos u + C = -\frac{1}{3} \cos(x^3) + C.$$

(iii) Let $u = 2x + 5$. Then $du = 2 dx \implies dx = \frac{1}{2} du$. Also, the upper limit $x = -2 \implies u = 2(-2) + 5 = 1$, and the lower limit $x = -5/2 \implies u = 2(-5/2) + 5 = 0$. So

$$\begin{aligned} \int_{-5/2}^{-2} x(2x+5)^8 dx &= \int_0^1 \frac{1}{2}(u-5)u^8 \frac{1}{2} du \\ &= \frac{1}{4} \int_0^1 u^9 - 5u^8 du \\ &= \frac{1}{4} \left[\frac{u^{10}}{10} - \frac{5u^9}{9} \right]_0^1 \\ &= \frac{1}{4} \left(\frac{1}{10} - \frac{5}{9} \right) \\ &= -\frac{41}{360}. \end{aligned}$$

(iv) Let $u = \sqrt{x}$. Then $du = \frac{1}{2\sqrt{x}} dx \implies \frac{1}{\sqrt{x}} dx = 2 du$. So

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int \sin u (2 du) = -2 \cos u + C = -2 \cos(\sqrt{x}) + C.$$

(v) Let $u = 1 + \tan t$. Then $du = \sec^2 t dt$. So

$$\int \frac{1}{\cos^2 t \sqrt{1 + \tan t}} dt = \int \frac{\sec^2 t}{\sqrt{1 + \tan t}} dt = \int \frac{1}{\sqrt{u}} du = 2\sqrt{u} + C = 2\sqrt{1 + \tan t} + C.$$

(vi) Let $u = \frac{t}{2}$. Then $du = \frac{1}{2} dt \implies dt = 2 du$. Also, the lower limit $t = \pi/3 \implies u = \frac{\pi/3}{2} = \pi/6$, and the upper limit $t = 2\pi/3 \implies u = \frac{2\pi/3}{2} = \pi/3$.

So

$$\begin{aligned}\int_{\pi/3}^{2\pi/3} \csc^2\left(\frac{t}{2}\right) dt &= \int_{\pi/6}^{\pi/3} \csc^2 u (2 du) \\&= 2 [-\cot u]_{\pi/6}^{\pi/3} \\&= 2 \left(-\cot \frac{\pi}{3} + \cot \frac{\pi}{6}\right) \\&= 2 \left(-\frac{1}{\sqrt{3}} + \sqrt{3}\right) \\&= \frac{4}{3}\sqrt{3}.\end{aligned}$$

□

5. If f is continuous on $[0, 1]$, show that

$$\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx.$$

Hint: Substitute $u = \pi - x$.

Hence, evaluate the integral:

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx.$$

You may assume that $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$.

Solution With hindsight, we will use the Substitution Rule:

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

which requires f to be continuous on the range of g .

In our case, $g(x)$ will be $\sin x$ (which is continuous on $[0, \pi]$), and so the range of $g(x) = \sin x$ is $[0, 1]$. Therefore, we will need to make the assumption that f is continuous on $[0, 1]$ if we were to use the Substitution Rule later on.

Let's make the substitution $u = \pi - x$. Then $du = -dx$, and the lower limit $x = 0 \implies u = \pi$, and the upper limit $x = \pi \implies 0$.

Also, note that

$$f(\sin x) = f(\sin(\pi - u)) = f(\sin u),$$

since $\sin(\pi - u) = \sin u$. By the Substitution Rule, we have

$$\begin{aligned} \int_0^\pi x f(\sin x) dx &= \int_\pi^0 (\pi - u) f(\sin u) (-du) \\ &= - \int_\pi^0 (\pi - u) f(\sin u) du \\ &= \int_0^\pi (\pi - u) f(\sin u) du \\ &= \pi \int_0^\pi f(\sin u) du - \int_0^\pi u f(\sin u) du \\ &= \pi \int_0^\pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx \\ &\quad \text{(by replacing the dummy variable } u \text{ by } x) \\ 2 \cdot \int_0^\pi x f(\sin x) dx &= \pi \int_0^\pi f(\sin x) dx \\ \int_0^\pi x f(\sin x) dx &= \frac{\pi}{2} \int_0^\pi f(\sin x) dx. \end{aligned}$$

This proves the first part of the question.

For the second part of the question, we need to find a function f such that $\frac{x \sin x}{1 + \cos^2 x} = x f(\sin x)$. Note that

$$\frac{x \sin x}{1 + \cos^2 x} = \frac{x \sin x}{1 + (1 - \sin^2 x)} = \frac{x \sin x}{2 - \sin^2 x} = x f(\sin x),$$

where

$$f(t) = \frac{t}{2 - t^2}.$$

Note that f is continuous on $[0, 1]$. By the above formula,

$$\begin{aligned}\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx &= \int_0^\pi x f(\sin x) dx \\ &= \frac{\pi}{2} \int_0^\pi f(\sin x) dx \\ &= \frac{\pi}{2} \int_0^\pi \frac{\sin x}{2 - \sin^2 x} dx \\ &= \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx.\end{aligned}$$

We proceed by substitution. Let $w = \cos x$. Then $dw = -\sin x dx$. Also, the lower limit $x = 0 \implies w = \cos 0 = 1$, and the upper limit $x = \pi \implies w = \cos \pi = -1$. Thus,

$$\begin{aligned}\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx \\ &= \frac{\pi}{2} \int_1^{-1} \frac{1}{1 + w^2} (-dw) \\ &= \frac{\pi}{2} \int_{-1}^1 \frac{1}{1 + w^2} dw \\ &= \frac{\pi}{2} [\tan^{-1} w]_{-1}^1 \\ &= \frac{\pi}{2} (\tan^{-1}(1) - \tan^{-1}(-1)) \\ &= \frac{\pi}{2} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) \\ &= \frac{\pi^2}{4}.\end{aligned}$$

□

6. Let $f(x) = \frac{1}{\sqrt{x}}$.

- (i) Find the average value of f on the interval $[1, 4]$.
- (ii) Find the value of c guaranteed by the Mean Value Theorem for Integrals such that $f_{ave} = f(c)$.

Solution

(i) The average value of f on $[1, 4]$ is

$$\frac{1}{4-1} \int_1^4 f(x) dx = \frac{1}{3} \int_1^4 x^{-1/2} dx = \frac{1}{3} [2\sqrt{x}]_1^4 = \frac{1}{3}(2\sqrt{4} - 2\sqrt{1}) = \frac{2}{3}.$$

(ii) We are looking for c such that

$$f(c) = \frac{2}{3} \implies \frac{1}{\sqrt{c}} = \frac{2}{3} \implies \sqrt{c} = \frac{3}{2} \implies c = \frac{9}{4}.$$

□

7. If $f_{ave}[a, b]$ denotes the average value of f on the interval $[a, b]$ and $a < c < b$, show that

$$f_{ave}[a, b] = \left(\frac{c-a}{b-a} \right) f_{ave}[a, c] + \left(\frac{b-c}{b-a} \right) f_{ave}[c, b].$$

Solution

$$\begin{aligned} f_{ave}[a, b] &= \frac{1}{b-a} \int_a^b f(x) dx \\ &= \frac{1}{b-a} \left(\int_a^c f(x) dx + \int_c^b f(x) dx \right) \\ &= \frac{1}{b-a} \int_a^c f(x) dx + \frac{1}{b-a} \int_c^b f(x) dx \\ &= \frac{c-a}{b-a} \left(\frac{1}{c-a} \int_a^c f(x) dx \right) + \frac{b-c}{b-a} \left(\frac{1}{b-c} \int_c^b f(x) dx \right) \\ &= \left(\frac{c-a}{b-a} \right) f_{ave}[a, c] + \left(\frac{b-c}{b-a} \right) f_{ave}[c, b]. \end{aligned}$$

□

Answer Keys. 1. (i) $\cos(x^2)$ (ii) $-\sqrt{1 + \sec s}$ (iii) $2 \left(-\frac{4y^2-1}{4y^2+1} \right) + 3 \left(\frac{9y^2-1}{9y^2+1} \right)$

2. (i) $-7/2$ (ii) 6 **3.** (i) $\frac{1}{5}f(5x)+C$ (ii) $\frac{1}{6}f(3x^2)+C$ **4.** (i) $-\frac{1}{3}(1-x^2)^{3/2}+C$
(ii) $-\frac{1}{3}\cos(x^3)+C$ (iii) $-\frac{41}{360}$ (iv) $-2\cos(\sqrt{x})+C$ (v) $2\sqrt{1+\tan t}+C$ (vi)
 $\frac{4}{3}\sqrt{3}$ **5.** $\frac{\pi^2}{4}$.
6. (i) $\frac{2}{3}$ (iii) $\frac{9}{4}$